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REVIEW

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Viral hemorrhagic fevers - therapeutic trial advances and challenges

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ABSTRACT

Introduction: Viral hemorrhagic fevers (VHFs) represent a group of severe diseases caused by RNA viruses with high fatality rates, posing significant global health threats. These diseases are prioritized by the World Health Organization due to their epidemic potential, geographical restrictions, and limited therapeutic options.

Areas covered: This review discusses the current state of therapeutic advancements, challenges in treatment, and post-exposure prophylaxis for various VHFs. Relevant literature on VHFs and therapeutic interventions was identified through searches of PubMed, Scopus, Web of Science, WHO and CDC databases, and ClinicalTrials.gov from database inception to November 2025. Key therapies like ribavirin for Crimean-Congo hemorrhagic fever and Lassa fever, along with monoclonal antibodies for Ebola and Marburg virus disease, have demonstrated clinical efficacy. However, gaps remain in effective antivirals and vaccines for many VHF pathogens.

Expert opinion: Notable challenges in therapeutic development include ethical concerns in randomized controlled trials, logistical barriers in endemic areas, and the evolving immune response in late-stage disease. The role of cytokine modulation and the growing potential of monoclonal antibodies offer new directions for treatment. Strengthening observational studies and expanding international collaboration are critical for improving patient outcomes and advancing therapeutic options for these deadly diseases.

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1. Introduction

Viral hemorrhagic fevers (VHFs) are caused by RNA viruses from several families, and are severe illnesses with a high fatality rate, posing serious global health threats due to their epidemic potential, geographic restriction, and limited treatment options. Many of these diseases are listed on the World Health Organization's (WHO) R&D Blueprint for priority pathogens – a designation that highlights both the urgent need for countermeasures and the persistent challenges in developing them, especially in low-resource settings where outbreaks typically occur [1]. Their inclusion reflects not only the unpredictability and biosafety concerns that hinder research but also the lack of commercial incentives, ethical barriers to randomized controlled trials, and fragility of healthcare systems in affected regions. In recent years, however, substantial progress has been made in developing and evaluating therapeutic candidates through innovative clinical trials and observational studies. This report reviews major therapeutic advances in VHFs and explores the key challenges that continue to shape the research and development landscape for these high-impact diseases.

A comprehensive narrative literature search was conducted to identify publications relevant to VHFs and their current or investigational therapeutic strategies. The search encompassed PubMed/MEDLINE, Scopus, Web of Science, WHO Global Health Library, CDC database, and ClinicalTrials.gov, covering studies published from database inception through November 2025. Search terms included combinations of 'viral hemorrhagic fever,' 'management,' 'antiviral therapy,' 'monoclonal antibody,' 'vaccine,' and disease-specific keywords (e.g. Ebola, Lassa, Dengue). Additional references were identified by screening relevant review articles and reference lists. Since this is a narrative (non-systematic) review, no formal inclusion/exclusion criteria or quality assessment tools were applied, and the literature search was updated continuously throughout manuscript preparation to ensure inclusion of the most recent studies.

2. General supportive care in VHFs

VHFs share a characteristic pathogenic cascade despite the diversity of causative viruses from different families [2]. Following viral entry into the host, the viruses infect mononuclear phagocytes and dendritic cells, dysregulated cytokine release and innate immune failures lead to endothelial cell activation and widespread vascular dysfunction [3,4]. The ensuing increased vascular permeability and plasma leakage contribute to hypotension and shock, while simultaneous platelet consumption, hepatic injury and consumption of clotting factors result in marked coagulopathy [5]. Together these processes culminate in multiorgan failure, hemorrhagic manifestations, and high case-fatality potential. This common pathophysiologic framework underpins the rationale for early, adequate supportive care and targeted therapies in VHF syndromes.

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Article highlights

- Viral hemorrhagic fevers (VHFs) share overlapping clinical and pathophysiologic features that guide common therapeutic and preventive strategies.
- Advances in antiviral and antibody-based therapies are transforming the management of VHFs, with several candidates showing protection in preclinical and clinical studies.
- Post-exposure prophylaxis and early therapeutic intervention remain critical to reducing mortality and limiting viral persistence in immune-privileged sites.
- Immune dysregulation, including cytokine overactivation, represents a key target for future adjunctive or immunomodulatory treatment strategies.
- Progress in VHF therapeutics is constrained by biosafety, ethical, and logistical challenges, underscoring the need for coordinated international clinical trial networks and equitable access to medical countermeasures.

Supportive care is the cornerstone of VHF management and has been shown to significantly improve survival across all types of these diseases [6]. Patients are typically managed in isolation units with strict infection control protocols, especially for the ones which human-to-human transmission risk is high. Key supportive measures include swift and adequate fluid resuscitation and electrolyte correction to address dehydration, third-spacing, and shock, often supplemented by vasopressors if hypotension persists [7]. Organ support such as oxygen therapy or mechanical ventilation may be necessary for respiratory failure, and dialysis may be indicated in cases of renal failure, with Intensive Care Unit (ICU) care required for multi-organ dysfunction [7]. Management of bleeding and coagulopathy is required for control of the disease. Even in the absence of visible bleeding, gastrointestinal or urogenital hemorrhages may occur. Blood products should not be administered empirically but based on clinical and laboratory assessments. In septic shock-like scenarios, red blood cell suspensions should be given to maintain a hematocrit level above 30% [8]. In locations where red blood cell suspension is unavailable, whole blood may be used [8]. Platelet transfusion should be applied when platelet count is < 20,000/µl in patients with bleeding, and < 10,000/µl in those without bleeding (1–2 units/10 kg) [9,10]. Fresh frozen plasma (FFP, 15–20 ml/kg) should be considered if fibrinogen levels are below 100 mg/dl. Although fibrinogen concentrates (total dose 2–3 g) or cryoprecipitates (1 unit/10 kg) may also be used, FFP is theoretically more advantageous as it contains all missing coagulation factors in DIC [11]. The blood products that can be used in hemorrhagic period is summarized in Table 1. Symptomatic treatments include antipyretics, analgesics, antiemetics, and agents for diarrhea management, while empirical antibiotic or antimalarial therapies are often administered to address potential co-infections in endemic regions [6,20]. Nutritional support is also vital, as patients typically experience high metabolic demands and reduced oral intake. Optimized supportive care protocols, developed by WHO and its partners, emphasize the importance of early intervention including prompt rehydration, maintenance of oxygenation, and treatment of complications which can dramatically improve outcomes even when specific antivirals are unavailable [21].

Despite advances in supportive care, most VHFs occur in low-to-middle income regions where access to medical equipment, ICUs, dialysis, and mechanical ventilation remains limited [7]. There is marked heterogeneity in healthcare settings, potentially driven by financial constraints, inadequate infrastructure, and physician-related factors, all of which shape the efficiency and consistency of supportive management provided to these patients [7]. During outbreaks, the scarcity of critical-care resources often necessitates simplified protocols emphasizing fluid resuscitation, electrolyte balance, oxygen supplementation,

Table 1. Blood products that can be used in the treatment of viral hemorrhagic fevers with indications, regimens and expected outcomes.

Blood Products	Primary indications	Standard regimen/considerations	Expected outcomes	References
Packed Red Blood Cells (PRBCs)	Severe anemia (Hb ≤6 g/dL), acute hemorrhagic shock with tissue hypoxia, increase oxygen-carrying capacity.	Dosage adjusted based on clinical response and ongoing/anticipated blood loss. Restrictive transfusion thresholds (Hb 7–8 g/dL) often supported.	Prevent tissue hypoxia. Rarely the primary cause of death is direct blood loss in VHFs.	[8,12,13]
Fresh Frozen Plasma (FFP)	Documented coagulation factor deficiencies, abnormal coagulation tests (prolonged aPTT) with active bleeding, liver disease, DIC.	Typically, 10–20 mL/kg (raises factor levels by ≈20%).	Restores coagulation factors. Contraindicated for simple volume expansion, protein supplementation, or coagulopathy without active bleeding/ impending procedure.	[8,13,14]
Platelets	Significant thrombocytopenia (PLT ≤20,000/µL is predictor of fatal CCHF outcomes) to prevent spontaneous bleeding or stop active hemorrhage.	Guided by careful clinical assessment of bleeding, not solely platelet counts. It can increase the platelet count by approximately 30,000–60,000/µL per apheresis unit or a pool of 4–6 units	Improves platelet count and hemostatic function. Routine prophylactic platelet transfusion is generally NOT recommended (e.g. in Dengue) due to limited efficacy and side effects.	[8,13,15–17]
Cryoprecipitate	Fibrinogen replacement (hypofibrinogenemia or dysfibrinogenemia) in actively bleeding patients or high hemorrhage risk.	Fibrinogen ≤2 g/L in life-threatening bleeding; ≤ 1 g/L with microvascular bleeding. Typical dose: 1 unit per 10 kg (approx. 10 units for adult) to increase fibrinogen by ≈0.5 g/L.	Provides fibrinogen, Factor VIII, vWF, Factor XIII.	[13,18,19]
Balanced Transfusion Strategy (PRBCs, FFP, Platelets)	Massive hemorrhage (loss of > 15% total blood volume or hemorrhagic shock).	Ratios typically 1:1:1 to 2:1:1.	Control life-threatening bleeding.	[8]

Abbreviations: Hb, Hemoglobin; aPTT, activated partial thromboplastin time; DIC, disseminated intravascular coagulation; PLT, platelets; CCHF, Crimean-Congo Hemorrhagic Fever; vWF, von Willebrand Factor.

and close hemodynamic monitoring using portable equipment [22]. Reinforcement of barrier nursing including strict use of personal protective equipment, patient cohorting, and adherence to safety protocol remains essential for the safe care delivery [23]. Strengthening local capacity for infection prevention, logistics, and training of frontline healthcare workers is essential to optimize outcomes in endemic settings [24].

3. Specific treatment and prevention measures

Across viral hemorrhagic fevers, the central therapeutic objective is early suppression of viral replication [22]. For many VHFs uncontrolled viremia correlates strongly with disease severity and mortality, as viral loads found to be higher in mortal cases [25–28]. Timely antiviral or neutralizing antibody administration is therefore crucial to curbing viral replication before the onset of immune dysregulation and irreversible tissue injury. This shared pathogenic feature underpins specific treatment strategies described below.

3.1. Ebola virus disease (EVD)

Ebola virus (Zaire ebolavirus) causes a highly lethal hemorrhagic fever. One of the largest outbreaks in West Africa in 2014 had fatality rate of 40% among 28 thousand patients [29], but survival has improved with modern care and new therapeutics. Supportive care remains essential, but EVD is unique among VHFs in having approved specific treatments [6]. Patients with EVD receive intensive supportive care similar to the general measures above. Although case fatality rates remain high, studies have suggested that outcomes may be more favorable in settings where comprehensive supportive care, including adequate fluid management, electrolyte monitoring, and organ support, is consistently available [8,30,31]. In addition, because Ebola often co-occurs with diseases like malaria in Africa, patients are treated for such co-infections as part of standard care [30]. ZMapp is a cocktail of three monoclonal antibodies (c13C6, c2G4, and c4G7) developed to treat EVD. It was designed to neutralize the Zaire ebolavirus glycoprotein and was tested during the 2014–2016 West Africa outbreak. Although initially used under compassionate protocols, ZMapp was formally evaluated in the PREVAIL II randomized controlled trial (RCT), which compared it to optimized supportive care alone. The trial showed a lower mortality rate in the ZMapp group (22%) compared to control (37%), but the difference was not statistically significant due to early termination and limited sample size [32]. Although the Zmapp cocktail was never approved, it was used as control group for upcoming RCTs of a novel monoclonal antibody (mAb) drug for treating EVD. In 2020, two mAbs treatments were approved after showing clear survival benefits in a large clinical trial. The PALM randomized trial in the Democratic Republic of Congo evaluated four investigational therapies for EVD including the mAb REGN-EB3, mAb114, the antibody cocktail ZMapp, and the nucleotide analogue remdesivir. The study found that the mAb REGN-EB3 and the mAb114 significantly reduced Ebola mortality compared to the ZMapp, while remdesivir did not show a statistically significant survival benefit, leading to its discontinuation within the study [33]. These neutralizing antibodies target Ebola virus glycoproteins, helping the immune system clear the infection. They are given as intravenous (IV) infusions, and when administered early in disease, they markedly improve survival. However, the PALM trial's inclusion of patients with advanced disease, often beyond the therapeutic window, may have confounded mortality comparisons, suggesting that the apparent inferiority of other agents including remdesivir and ZMapp, should be interpreted with caution [34]. Moreover, the recent study using a uniformly lethal rhesus-monkey model demonstrated that when treatment was initiated five days post-exposure, all four agents achieved similar survival rates, with survival closely linked to viral load at treatment onset [35]. These data imply that the apparent inferiority of remdesivir and ZMapp in the PALM trial may reflect timing of intervention rather than intrinsic lack of efficacy. Despite contradictory findings, WHO issued strong recommendations to treat EVD with either mAb114 or the REGN-EB3 cocktail [36]. Beyond mAbs, favipiravir was tried in earlier outbreaks as an experimental therapy; however, data on its efficacy in Ebola are limited [37–39]. Research continues into treatments for other Ebola species (like Sudan virus and Bundibugyo virus), which currently have no approved therapeutics. During a 2022 Sudan Ebola outbreak in Uganda, investigational mAbs and antivirals were deployed under trial protocols WHO and partners have developed a 'core' clinical trial protocol to evaluate such candidate treatments during filovirus outbreaks [40]. Combination regimen of remdesivir with monoclonal antibodies, which have shown synergistic efficacy and improved survival in non-human primate models [41], was also used in this outbreak. The rationale behind such combination therapy is to enhance antiviral efficiency by targeting different viral mechanisms simultaneously, thereby shortening the therapeutic window for effective intervention. In parallel, novel broad-spectrum antivirals are being developed. Among them, obeldesivir, a next-generation nucleoside analogue, has demonstrated robust efficacy against SUDV and EBOV infections in non-human primates, significantly improving survival when administered early [42,43].

Vaccine development for EVD has progressed substantially over the past decade, resulting in two licensed vaccines for Zaire EBOV and several promising next-generation candidates. The rVSV-ZEBOV vaccine, a live recombinant vesicular stomatitis virus expressing the Ebola virus glycoprotein, demonstrated high efficacy in a ring-vaccination trial during the 2015 West African outbreak [44]. The heterologous Ad26.ZEBOV/MVA-BN-Filo prime-boost regimen has also been licensed and shown to elicit durable humoral and cellular immune responses in both adults and children, supporting its use for preventive immunization in high-risk regions [45,46]. Additional adenoviral-based constructs, including ChAd3-EBOZ, continue in clinical evaluation, while multivalent platforms aiming to protect against multiple filoviruses are advancing through early-phase trials [47]. Together, these candidates reflect the maturation of Ebola vaccine research from emergency deployment to broader preventive and multivalent approaches.

EVD survivors frequently suffer from Post-Ebola Syndrome (PES), which includes severe arthralgias, chronic fatigue,

headaches, hearing loss, and potentially vision-threatening uveitis [48,49]. Treatment is symptomatic and rehabilitative, involving ophthalmologic and audiological care for sensory loss, physical therapy for joint and muscle pain, and mental health support for the common psychological sequelae like depression [49]. Persistent virus in immune-privileged sites requires ongoing monitoring to prevent recurrence and sexual transmission [50].

3.2. Marburg virus disease (MVD)

Marburg virus is a filovirus closely related to Ebola, and Marburg virus disease causes similar clinical syndrome with high fatality rate of 61% in a metanalysis [51]. No specific antiviral treatment or vaccine is yet approved for MVD, making supportive care the primary therapy [52]. Patients are managed in isolation with rigorous supportive measures: rehydration, hemodynamic support, and management of bleeding and organ failure, as described above. Early intensive supportive care improves survival chances in Marburg infection [53].

Despite the critical need, no antiviral drugs are approved for MVD. However, promising research highlights several candidates. Oral obeldesivir has shown life-saving potential in non-human primate models, demonstrating 80% survival when administered post-exposure, by reducing viral replication and delaying disease onset [54]. Favipiravir, galidesivir, and remdesivir are under investigation for potential benefit [55].

New therapeutic strategies are also exploring mAbs and host-targeting antivirals. Several mAbs are being developed to specifically neutralize Marburg virus, with some demonstrating cross-reactivity with other filoviruses [56]. MBP091 is derived from a human survivor's antibody and targets the Marburg virus glycoprotein to neutralize the virus. Preclinical studies suggest that combining MBP091 with remdesivir may synergistically improve survival, extending the window of effective treatment [53,57]. Additionally, research is delving into host-targeting antivirals, including immune modulators like interferons and cytokine/chemokine modulators, which aim to correct the immune dysregulation seen in MVD [55,58]. While no Marburg vaccine is yet approved, several candidates are in advanced clinical development. The WHO has highlighted promising platforms, including chimpanzee adenovirus (cAd3) vector and recombinant vesicular stomatitis virus-based vaccines [52,59]. The Sabin Vaccine Institute is conducting Phase 2 trials for its cAd3-based candidate in the U.S., Uganda, and Kenya, evaluating safety and immunogenicity [60]. These vaccines have shown protective efficacy in preclinical models, but challenges like sporadic outbreaks and logistical difficulties in deployment remain [59,61].

Long-term sequelae in MVD are less extensively studied than EVD, but survivors often exhibit persistent cardiometabolic alterations, subclinical hepatic/renal stress, and long-term hematological modulation resulting in reticulocytosis [62]. Psychological morbidity is reported to be less marked compared to EVD, but general supportive care is needed for lingering fatigue and aches [62]. Therapeutic options involve routine chronic disease management such as blood pressure monitoring.

3.3. Lassa fever

Lassa fever, caused by an arenavirus, is endemic in West Africa (hundreds of thousands of infections annually) with a lower overall fatality (~1% overall, but up to 15% in hospitalized severe cases). However, Nigeria reported an overall fatality rate ranging from 15.4% to 18.5% in 2025 [63]. The mainstay of treatment is supportive care, as no specific antiviral is formally approved [64]. Key supportive measures include fluids, blood pressure support, and managing complications like renal failure or neurological symptoms. Early recognition and hospitalization improve outcomes, as does care in specialized centers that can provide isolation and intensive monitoring.

Since the 1980s, the antiviral ribavirin has been used off-label as the de facto treatment for Lassa fever. If given early in the course (within the first 6–7 days of illness), ribavirin has been reported to reduce mortality, though the evidence is based on observational studies [65,66]. While clinical data have long suggested some therapeutic benefit, recent meta-analysis have challenged this assumption, revealing inadequate antiviral activity in preclinical studies [67]. This discrepancy underscores the urgent need for well-validated, unequivocally effective therapeutics against Lassa fever.

Recognizing the need for better treatment, a major effort is underway to test new antivirals and immunotherapies for Lassa fever. In 2024, the *INTEGRATE consortium* (a global alliance of researchers and NGOs) initiated the first large-scale Lassa fever treatment trials in West Africa in nearly 40 years [68]. These adaptive trials, starting in Nigeria and planned for Liberia and Guinea, will compare ribavirin with newer drug candidates and allow dropping ineffective arms quickly. The goal is to find more effective drugs to treat Lassa by testing antivirals used for other diseases or novel agents. Candidates likely to be evaluated include favipiravir, which has shown in-vitro activity against Lassa and was assessed in a small Phase II trial [69], ST-193 (an experimental Lassa antiviral), and possibly mAbs. Among those, the Arevirumab-3 cocktail of three broadly neutralizing human mAbs has been developed over more than a decade and represents the first LASV-specific therapeutic with robust post-exposure efficacy in non-human primates, rescuing up to 100% of macaques from otherwise lethal infection when treatment is initiated at advanced stages of disease and across multiple lineages that have caused human disease [70–72]. Consequently, Arevirumab-3 is increasingly viewed as a promising advancement in Lassa therapeutics and a potential candidate for future clinical evaluation.

Supportive care improvements in Lassa are also being emphasized. For example, managing Lassa's frequent complication of hearing loss with early steroids or other interventions is an area of research though evidence is limited [73]. Because Lassa can cause fluid overload issues and shock, careful fluid management and treatment of secondary bacterial infections can make a difference in outcome. Some severe Lassa patients require dialysis due to acute renal injury or ventilation for respiratory failure; having these capabilities in endemic-area hospitals is part of regional preparedness.

Lassa Fever is notorious for causing permanent, often bilateral sensorineural hearing loss which often coexists with balance difficulties, seizures, and cognitive impairment [74]. Other sequelae include chronic renal dysfunction, ophthalmic complications such as cataracts, and psychological disorders [74,75]. Management is multidisciplinary and rehabilitative, including audiology, specialized renal care, ophthalmic intervention, and dedicated mental health and psychosocial support.

3.4. Crimean-congo hemorrhagic fever (CCHF)

CCHF is a tick-borne VHF endemic in parts of Africa, the Middle East, Asia, and Eastern Europe, including countries such as Türkiye, Bulgaria, and Spain [20]. It presents with high fever, bleeding, and often shock, with a mortality rate generally estimated between 10% and 40% by the WHO [20]. However, mortality rates vary significantly by region; for example, Türkiye, which reports the highest number of global cases, has observed a case fatality rate of around 5% [76]. In contrast, data from the European Centre for Disease Prevention and Control covering confirmed cases in the EU/EEA since 2013 show an overall case fatality rate of approximately 23%, within 65 cases in total [77]. Currently, there is no specific antiviral treatment or approved vaccine for CCHF. Management remains primarily supportive, requiring strict isolation precautions due to the risk of transmission via blood and body fluids.

Ribavirin remains the most studied and widely used antiviral agent for the treatment of CCHF. Its efficacy has been supported by *in vitro* studies, animal models, and multiple observational clinical studies, particularly when administered during the early, viremic phase of illness [78–81]. Although placebo-controlled RCTs are often ethically challenging due to the high mortality of the disease, retrospective analyses, such as a large-scale study from Turkey, have demonstrated significant reductions in fatality rates with early ribavirin use, even after adjusting for disease severity [82]. Similarly, early ribavirin administration, before 96 hours after symptom onset, was associated with lower mortality rate [83]. Ribavirin is generally well tolerated in CCHF patients and remains recommended by international authorities such as WHO, ECDC, and CDC for both treatment and post-exposure prophylaxis [20]. Despite a lack of definitive placebo-controlled trial data, expert consensus and clinical experience strongly support its use. A study by US Army Medical Research and Development Command which compare IV ribavirin to standard of care in CCHF and Lassa fever was reported to continue, however there is no data shared (NCT02483260).Emerging treatments for CCHF include favipiravir, which has demonstrated efficacy in immunocompetent animal models [80]. Basic research has identified new antiviral targets in the viral lifecycle, such as the subtilisin kexin isozyme-1/site-1 protease [84] and the low-density lipoprotein receptor, a recently discovered entry receptor for the virus [85]. Additionally, high-throughput screening is uncovering novel compounds with in vitro antiviral activity [86,87]. Research on mAbs derived from human survivors of CCHF has identified potent neutralizing antibodies binding to the viral glycoprotein Gc, as well as an engineered bispecific antibody that provided therapeutic protection in mice even when administered post-exposure, promising future antibody-dependent treatment of CCHF patients [88].

As CCHF progresses beyond the initial viremic phase, the role of antivirals like ribavirin diminishes significantly. Once viremia subsides, typically around the onset of hemorrhagic manifestations, the pathophysiology shifts from direct viral replication to an exaggerated host immune response, including cytokine storm and endothelial dysfunction. In this late phase, antiviral agents are unlikely to provide therapeutic benefit, and disease severity is often driven by immune-mediated damage rather than viral load [3]. Consequently, immunomodulatory therapies such as corticosteroids, anti-TNF-α agents, nitric oxide inhibitors, or other cytokine-targeting interventions have been proposed as potential adjuncts [89]. Although their efficacy remains unproven and investigational, they may hold promise in mitigating the immune-driven pathology characteristic of late-stage CCHF. Understanding the timing of therapeutic intervention is therefore crucial: early antiviral use targets viral replication, while late-phase strategies may need to focus on modulating the host response.

Despite CCHF is endemic across extensive regions of Europe, Asia, and Africa, most therapeutic clinical trial activity has been concentrated in a limited number of endemic countries and in locations where candidate products are being developed, particularly in the context of vaccine research. The absence of published results from completed or withdrawn studies may partly explain the cautious and adaptive design adopted in ongoing clinical trials. The forthcoming UMIT-2 Phase IIb platform trial in Türkiye aims to systematically evaluate favipiravir and ribavirin against optimized standard of care, reflecting an effort to generate more definitive efficacy data through standardized endpoints and harmonized clinical protocols (NCT06860334). Given that Türkiye reports the highest number of CCHF cases annually, it remains a key hub for clinical research, yet expansion of well-structured trials into underrepresented African endemic regions is essential to ensure globally representative data and equitable access to future therapeutics.

A well-established, clinically proven therapeutic for CCHF is urgently needed, supported by robust phase 2 and 3 studies to ensure efficacy across different virus genotypes. To achieve statistically meaningful outcomes, active participation from all endemic countries and a coordinated candidate prioritization process are essential [86]. Given the substantial time, infrastructure, and resources required to develop new antivirals, repurposing existing licensed drugs offers a more feasible and strategic approach, particularly as a short-term solution while novel treatments continue through early-stage development.

3.5. Dengue virus (DENV)

Dengue, caused by any of the four dengue virus serotypes (DENV-1 to DENV-4) within the *Flaviviridae* family, can result in a spectrum of illness ranging from mild febrile episodes to severe manifestations such as dengue hemorrhagic fever and dengue shock syndrome, characterized by plasma leakage, hemorrhage, and organ impairment [90]. Currently, there is no specific antiviral treatment approved for dengue virus

infection. Management remains entirely supportive, with careful and judicious fluid replacement being the cornerstone of therapy, especially during the critical phase when vascular leakage occurs, typically between days 3 and 7 of illness [91]. This phase requires close monitoring of hemodynamic status and hematocrit levels to promptly identify signs of hypovolemia and guide fluid resuscitation. Adjunctive therapies such as antifibrinolytic agents have been explored; tranexamic acid, for instance, has been studied in small trials for its potential to reduce bleeding, but results have not demonstrated consistent clinical benefit, and its use is not currently recommended in standard dengue management protocols [15].

Multiple antiviral candidates, including celgosivir and balapiravir, have been evaluated in past clinical trials; however, both failed to demonstrate significant virological or clinical efficacy in dengue patients [92,93]. More recent preclinical investigations have focused on agents like favipiravir which shown to has strong binding affinity to the NS5 polymerase complex of all four DENV serotypes in molecular docking and simulation studies, but clinical trials are still required to assess its real-world efficacy [94].

Immune responses play a critical role in dengue pathogenesis. Early innate immune activation involves interferon production, but dengue viral proteins (e.g. NS5, NS4A) actively suppress these pathways, facilitating immune evasion and enhanced replication [95]. Secondary infections often lead to antibody-dependent enhancement (ADE), where non-neutralizing antibodies promote viral uptake via Fcγ receptors, contributing to severe disease manifestations like dengue hemorrhagic fever and shock [91]. Due to the immunopathological nature of severe dengue, corticosteroids have been investigated as potential immunomodulatory treatments. However, clinical trials and observational studies have produced inconclusive results [96–99]. Consequently, corticosteroids are not recommended in current WHO guidelines, and their routine use remains controversial [90].

As no antiviral drug has yet reached the threshold for regulatory approval, prevention remains the most effective strategy to combat dengue. This includes integrated vector management, public health education, and immunization. Of the vaccines developed, two live-attenuated tetravalent vaccines have reached licensure. The first, CYD-TDV (Dengvaxia) demonstrated partial efficacy and remains the first licensed dengue vaccine [100]. However, its use is restricted to individuals with documented prior dengue infection, as post-licensure surveillance revealed an increased risk of severe disease among seronegative recipients [101]. Despite these limitations, CYD-TDV represents an important milestone in dengue prevention and informed the design of subsequent candidates. The second, TAK-003 (QDenga), a live-attenuated tetravalent vaccine, has shown efficacy against virologically confirmed dengue and has been approved in several countries [102,103]. However, its use is currently recommended primarily for children aged 6 to 16 years in high-transmission settings, due to limited efficacy data in seronegative individuals and certain serotypes [90]. Ongoing trials are evaluating the long-term safety and efficacy of TAK-003 across diverse age groups and geographic regions [104], as well as many vaccine trials are being conducted within different phases [105]. Until a safe, effective, and widely applicable antiviral or vaccine becomes available, dengue control will continue to rely on a combination of supportive care and preventive public health measures.

Survivors of dengue including those with non-severe acute illness may experience a post-dengue syndrome (PDS) characterized by prolonged fatigue, myalgia or arthralgia, and cognitive difficulties such as impaired concentration [106,107]. Emerging population-based data further indicate that dengue infection can lead to long-term, multi-systemic complications beyond the acute phase, underscoring the need for post-recovery follow-up and multidisciplinary management of affected individuals [108]. Treatment for PDS is symptomatic and relies on supportive care such as pain management, physical therapy, and psychosocial support. The therapeutic focus is on gradually restoring the patient's functional abilities and quality of life over months.

3.6. Yellow fever (YF)

YF is an acute viral hemorrhagic disease transmitted primarily by infected *Aedes aegypti* mosquitoes, endemic to tropical and subtropical regions of Africa and South America [109]. The clinical presentation of YF is highly variable, ranging from mild, flu-like symptoms to a severe, life-threatening illness characterized by liver damage, bleeding, and jaundice [110]. The disease's name itself reflects the characteristic yellowing of the skin and eyes observed in severe cases.

The yellow fever vaccine 17D is a highly successful live-attenuated vaccine, first introduced in 1937, which has been instrumental in controlling yellow fever virus globally [111]. It is widely recommended for travelers to yellow fever risk areas as a primary preventive measure against the disease [110]. Its widespread use has significantly reduced the global burden of YF. Despite its long-standing success in the field, there are 773 confirmed cases of YF reported until 2023 for 79 years [111]. Ongoing research, such as a phase IV clinical trial (NCT01290055), has recently completed the study about the immune responses to the 17D vaccine, focusing on the processing and lifespan of YF-specific CD8+ T cells, aiming to deepen the understanding of vaccine-induced immunity [112]. A significant limitation of the vaccine in controlling active outbreaks is the time required to induce protective neutralizing antibodies, which can take weeks, particularly in elderly individuals. This delay means that vaccination efforts during ongoing epidemics may be ineffective to protect from infection leading to cases of individuals becoming infected and hospitalized even after recent vaccination [113,114].

Despite extensive research into antiviral therapies for other members of the *Flaviviridae* family, such as Hepatitis C virus and DENV, no approved antiviral drugs specifically targeting YF virus have been developed or are available for human use [115]. This suggests unique challenges in targeting YFV's viral proteins or replication mechanisms, or a comparative lack of focused investment for this particular flavivirus. Experimental therapies such as ribavirin and interferon have been explored in certain cases or animal models, but their reliance on high

doses administered shortly after infection limits their feasibility for clinical application [116,117].

Recent advances in YFV therapeutics have focused on mAbs as potential therapeutic agents. A Phase 1a/b clinical trial (NCT03776786) evaluated the safety, pharmacokinetics, and efficacy of the intravenously administered human mAb TY014, the first YFV-specific antibody candidate to reach clinical trials. The infusions were well tolerated and demonstrated suppression of viremia following vaccination with the attenuated YFV-17D strain, although the study was limited by its reliance on a vaccine-challenge model rather than wild-type infection [118]. Additional preclinical studies in rodent and non-human primate models have shown that YFV-specific antibodies, particularly MBL-YFV-1 and MBL-YFV-2, can prevent severe disease and death, supporting their potential for therapeutic or post-exposure use [114]. Despite these encouraging results, clinical data remain scarce, and further studies are needed to confirm efficacy in natural YFV infection and to define their optimal role alongside vaccination strategies.

A major obstacle in advancing and implementing YFV antiviral therapies lies in overcoming economic and accessibility barriers; even with an effective drug, its public health impact will depend on affordability, equitable distribution, and integration into healthcare systems particularly in regions with low vaccine uptake, underscoring the need for therapeutic development to be coupled with comprehensive access strategies.

3.7. Rift valley fever (RVF)

RVF is a zoonotic viral disease caused by the RVF virus, primarily found in most of sub-Saharan Africa, particularly eastern and southern Africa. The virus is transmitted to humans through the bite of infected mosquitoes, but also significantly through direct contact with the blood, body fluids, or tissues of infected animals, such as during slaughtering, butchering, or veterinary procedures [119]. The primary treatment for RVF remains supportive care, as most cases are mild and resolve spontaneously [120]. Over-the-counter medications like ibuprofen or acetaminophen can help manage symptoms of mild illness, such as fever and body aches.

Currently, there are no approved or effective antiviral drugs for human use specifically targeting RVF. In investigational studies, ribavirin showed a delay in disease onset in RVFV-infected STAT-1 KO mice, but its overall clinical benefit in humans for RVF remains unestablished and its adverse effect profile is a concern [121,122]. A promising development involves the use of artificial intelligence-based virtual high-throughput screening. This approach successfully identified several small molecule compounds targeting the Gc fusion mechanism of RVFV. Three out of 94 selected compounds effectively inhibited RVFV replication in Vero cells when administered as a pretreatment, indicating an inhibitory effect upon viral entry [123].

Research on mAbs against RVFV is advancing, with efforts focused on generating and characterizing antibodies targeting key structural proteins such as the nucleoprotein, Gc, and Gn, using advanced techniques like high-throughput single-cell analysis to identify rare and potent candidates [124,125]. This detailed epitope mapping enables the development of highly specific therapies that can disrupt critical viral functions, with 23 recombinant mAbs already identified and characterized for their antigen-binding properties, offering promising avenues for both therapeutic innovation and mechanistic insight [126].

There are currently no licensed vaccines for RVF available for human use. While veterinary vaccines are used in endemic regions to control outbreaks in livestock, the absence of a human vaccine highlights a critical gap in global health preparedness [127]. Several human vaccine candidates are undergoing clinical evaluation, including both inactivated and live-attenuated platforms, which have shown promising safety and immunogenicity profiles in early-phase trials [127,128]. The ChAdOx1 RVF vaccine, based on a chimpanzee adenoviral vector encoding the RVF virus glycoproteins, is being assessed in Kenyan adults to determine its safety and immunogenicity (NCT06799234). The trial aims to establish whether this vector platform elicits a protective immune response suitable for endemic populations. In parallel, the TSI-GSD 200 inactivated vaccine (Lot 7, Run 2) is being evaluated among laboratory and field personnel who may be occupationally exposed to RVF virus, providing updated safety and immunogenicity data for this legacy vaccine formulation (NCT03609398). Together, these studies represent important progress toward defining safe and effective vaccines for both endemic and high-risk occupational groups.

The most devastating sequelae of RVFV are highly specific: ocular disease leading to permanent partial or complete blindness and meningoencephalitis leading to long-lasting neurological deficits [119,129]. There is no specific treatment for the virus after the acute phase, so therapy for sequelae is specialized and rehabilitative; this includes ophthalmological intervention for the retinal lesions and specialized neurological rehabilitation for the encephalitis-related damage.

3.8. Hantavirus

Hantaviruses are rodent-borne bunyaviruses that cause two distinct syndromes: Hemorrhagic Fever with Renal Syndrome (HFRS) in Asia and Europe, and Hantavirus Pulmonary Syndrome (HPS) in the Americas. HFRS presents with fever, headache, back pain, and can progress to vascular leakage, shock, and acute kidney injury [130]. HPS is characterized by a rapid onset of respiratory distress, leading to acute lung edema and cardiogenic shock, with high fatality rates [130]. For both syndromes, intensive supportive care in an ICU setting is fundamental. This includes renal dialysis for HFRS, where kidney failure is prominent, and extracorporeal membrane oxygenation or mechanical ventilation for HPS, which causes severe acute lung edema [131].

In terms of antiviral treatment, IV ribavirin has been evaluated for HFRS in RCTs in China and Korea, demonstrating a significant reduction in mortality, particularly when administered early in the disease course [132,133]. However, other studies including a RCT on ribavirin for HPS have failed to show a significant benefit [134,135]. This lack of efficacy in HPS may be attributed to the fact that most patients had already

developed advanced cardiopulmonary edema by the time treatment began, underscoring the critical importance of timing in antiviral intervention for hantaviruses [134].

Beyond ribavirin, research into other antivirals and novel therapeutic strategies for hantaviruses is ongoing, but none are yet available for clinical use. Favipiravir has shown promising in vitro activity against hantaviruses and efficacy in lethal hamster models for HPS-causing viruses like Sin Nombre virus and Andes virus (ANDV) [136]. Furthermore, a human mAb has been discovered that effectively neutralizes both Old World and New World hantaviruses in animal models, offering a promising candidate for a broad-spectrum passive immunotherapy [137]. However, these remain in the research and development pipeline, without current clinical application.

Prevention through vaccination is a key strategy, particularly for HFRS. Effective inactivated vaccines are utilized in China and Korea for HFRS prevention, although these are prophylactic measures rather than treatments for active infection [138]. For HPS, there is no widely licensed vaccine currently available, though various platforms including subunit, viral vector, and nucleic acid vaccines are under extensive research [131]. The development of broadly protective vaccines and effective, clinically available antiviral treatments remains a significant challenge for both HFRS and HPS, particularly given the rapid progression of severe forms of the diseases.

Hantavirus infection often leads to chronic renal or pulmonary issues depending on the strain. Survivors of HPS frequently report long-term shortness of breath, decreased exercise tolerance, and fatigue due to mild to moderate pulmonary changes [139]. Survivors of HFRS may develop chronic proteinuria and progressive renal insufficiency that can worsen over years [140,141]. Treatment focuses on chronic disease management, which involves long-term nephrology follow-up for renal function, and pulmonary rehabilitation to improve exercise capacity and breathing.

3.9. Arenaviruses

Arenaviruses, including Junín virus (JUNV) responsible for Argentine Hemorrhagic Fever (AHF) and Machupo virus (MACV) causing Bolivian Hemorrhagic Fever (BHF), are rodent-borne viruses that can cause severe hemorrhagic fevers akin to Lassa fever [142]. For both, supportive care, addressing symptoms like fever, pain, and maintaining fluid and electrolyte balance, is crucial.

A notable success story in arenavirus treatment is the use of convalescent plasma for AHF. When administered within 8 days of symptom onset, plasma from recovered patients dramatically reduces JUNV mortality, from 20–30% down to less than 3% [143]. This highly effective passive immunotherapy has been a standard treatment in Argentina for decades, profoundly impacting disease outcomes. While ribavirin has shown some *in vitro* activity against arenaviruses, and has been explored for late-stage AHF, it has not demonstrated the same significant mortality reduction as immune plasma [144–146]. Other novel antivirals, including modified ribavirin and favipiravir have shown promise in preclinical studies against JNV, but their clinical utility is still under investigation [144,145,147].

Beyond convalescent plasma, research into new drugs for arenavirus infections includes the development of mAbs. These are targeting arenavirus glycoproteins, aiming to provide specific neutralizing activity and potentially enhance host immune responses [148]. For Machupo virus, polyclonal antibodies targeting glycoproteins have shown protection in guinea pig models [149], and specific mAbs against MACV have shown neutralizing activity and Fc-effector functions *in vitro* and in mouse models [150]. Similarly, mAbs directed against Junín virus (J199, J200, and J202) have provided complete protection in non-human primates [151]. Immunomodulators beyond the direct effects of immune plasma or mAbs are not standard specific treatments for these arenaviruses.

In terms of prevention, Argentina developed a live-attenuated JNV vaccine, Candid #1, which has been highly successful and has virtually eliminated AHF in endemic areas [152]. For other New World arenavirus hemorrhagic fevers, such as those caused by MACV, no licensed human vaccine is currently available. However, a recombinant MACV expressing Candid #1 glycoprotein is a promising vaccine candidate in mouse models [153].

4. Post-exposure prophylaxis (PEP) for VHFs

PEP for VHFs plays a critical role in preventing disease progression following exposure to high-risk environments, particularly for healthcare workers (HCWs) and individuals in close contact with infected patients. Importantly, PEP strategies may also reduce viral persistence in immune-privileged sites such as the eyes, central nervous system, and testes, which are recognized reservoirs for recrudescence and secondary transmission in filovirus and arenavirus survivors [154,155]. Among the most well-established PEP strategies is the use of the rVSV-ZEBOV vaccine for EVD which has demonstrated high efficacy when administered within 72 hours of exposure [156]. For Marburg virus, mAbs such as MR191-N have proven highly protective in animal models when given shortly after exposure [55]. Moreover, obeldesivir has shown potential in post-exposure treatment for both Ebola and Marburg viruses, providing protection when administered within 24 hours of exposure in preclinical trials [54], and there is a planned Phase II study (NCT06682234).

Ribavirin has emerged as a recommended option for PEP against certain VHFs, particularly CCHF and Lassa fever. Evidence from systematic reviews and meta-analyses indicates that ribavirin significantly reduces the risk of infection among healthcare workers after high-risk exposure to CCHF patients, with odds ratios as low as 0.01, and early administration within 48 or 96 hours after symptom onset markedly decreases mortality [83,157]. In the case of CCHF, ribavirin is widely used for post-exposure prophylaxis, demonstrating effectiveness in reducing the risk of infection when administered promptly. A similar role is observed for Lassa fever, although no prospective trials exist, clinical experience and limited observational data suggest that oral ribavirin may be reasonable for PEP after confirmed high-risk exposures such as needlestick injuries or direct mucosal contact with infectious fluids [66].

However, adherence challenges and adverse effects have been noted, as described in a Sierra Leone outbreak, where less than half of exposed individuals completed a ribavirin regimen due to side effects, though no secondary infections were confirmed [158]. Given its tolerability profile, ribavirin may not be suitable for large-scale or community-level PEP (e.g. after general tick exposure), but it remains an appropriate and evidence-supported option for targeted prophylaxis among healthcare and laboratory personnel following verified high-risk exposure. The continued development and evaluation of possible PEP agents will have a key role in improving outcomes for individuals exposed to high-risk environments and enhancing public health preparedness for future outbreaks.

5. Challenges in developing therapeutics

Supportive care remains the cornerstone of treatment for VHFs, requiring strict clinical monitoring and ICU admission when indicated. Nonetheless, early therapeutic intervention during the viremic phase has shown potential to reduce mortality, exemplified by ribavirin use in CCHF and Lassa fever. The deployment of orally available antivirals represents a crucial advancement for managing VHFs in both outbreak and endemic contexts. Oral administration minimizes the need for parenteral access, thereby reducing occupational exposure risk to healthcare workers and mitigating further vascular injury in patients with coagulopathy or thrombocytopenia [159]. Early oral initiation also facilitates treatment outside intensive-care settings, an advantage during large-scale outbreaks when intravenous administration is logistically constrained.

The design of RCTs incorporating placebo arms is often constrained by ethical and practical challenges, particularly in high-mortality settings. Moreover, VHFs are largely endemic to resource-limited regions of Africa and Asia, which restricts the feasibility and external validity of standalone RCTs without significant international collaboration. In these contexts, withholding potentially life-saving interventions in placebo-controlled trials raises profound ethical dilemmas, reinforcing the relevance of observational, in vitro, and in vivo studies, especially when performed within endemic populations where real-world disease behavior can be captured. Current treatment strategies supported by robust observational data and available clinical trials are outlined in Table 2.

While well-designed observational studies are critical, their methodologies must account for the shifting pathogenesis of VHFs, which typically evolve from a viremic phase to an immune-mediated stage dominated by the host response. Secondary infections, as commonly seen in dengue, may further complicate patient management and data interpretation. Observational research must carefully address variables such as disease severity, treatment timing, and clinical stage to minimize bias. One key source of bias is confounding by indication, where patients with more severe illness are more likely to receive aggressive therapy, potentially skewing perceptions of treatment effectiveness due to higher baseline mortality. Furthermore, as antivirals are only effective during active viremia, documenting precise treatment initiation and duration is essential to accurately interpret outcomes and guide future research. Figure 1 illustrates the typical disease course of viral hemorrhagic fevers following exposure, alongside recommended therapeutic interventions tailored to each phase.

In VHFs, a surge in cytokine levels, often referred to as a 'cytokine storm,' plays a critical role in the pathogenesis of the disease. Elevated levels of pro-inflammatory cytokines, such as IL-1, IL-6, and IFN-γ, and others, are commonly observed in patients with severe forms of these infections [160–163]. This cytokine surge is associated with increased disease severity and poor clinical outcomes, making cytokine pathways a potential target for therapeutic intervention [10]. Targeting specific cytokine pathways, particularly IL-6 and TNF-α, could offer promising therapeutic strategies for managing VHFs. However, the use of general immunosuppressants, such as corticosteroids, remains controversial and poorly understood in the context of VHFs. While corticosteroids have been used in certain cases, their efficacy has not been conclusively proven [82]. Further studies are needed to clarify the role of immunomodulatory treatments, including interleukin inhibitors and corticosteroids, in the management of VHFs.

mAbs are emerging as potential therapeutics for VHFs, with proven efficacy in EVD while others are still under investigation. However, one significant challenge in their application is the requirement for intravenous infusion, which necessitates that mAbs be administered in hospital settings. This requirement presents complications, particularly when patients with mild symptoms are confirmed with VHFs and hospitalized. The risk of healthcare-associated infections increases in these hospital environments, complicating the overall treatment strategy.

Simultaneously, many vaccine studies are being conducted for various VHFs, progressing through different phases of clinical trials. Vaccines are undeniably critical to preventing further outbreaks, but their production, distribution, and access present substantial challenges. Since most VHFs are endemic in low- to middle-income countries, these regions face difficulties affording the costs associated with transporting and administering vaccines, necessitating support from international organizations. Another key issue is determining how to prioritize risk groups for the first doses, particularly when VHFs have the potential to spread to new areas, as seen with CCHF in Europe. This raises the question of whether high-income countries should receive vaccines first, or if the priority should be given to populations living in endemic regions.

A further barrier to the rapid development of medical countermeasures against VHFs is the requirement for biosafety level 4 (BSL-4) containment for nearly all preclinical efficacy studies. Only a small number of laboratories worldwide are equipped to handle these pathogens, and access is limited by stringent regulatory oversight, high operational costs, and intensive personnel training requirements [25]. These factors substantially slow the evaluation of candidate antivirals and vaccines, delaying translation of promising laboratory findings into clinical studies [25]. Expansion of collaborative BSL-4 capacity and establishment of surrogate or pseudo virus models under lower containment may help accelerate early-stage research.

Table 2. Current therapeutic options for viral hemorrhagic fevers, based on most relevant clinical studies.

Viral hemorrhagic fever	Antivirals			Immunomodulator drugs		Monoclonal antibodies
	Ribavirin	Favipiravir	Other antivirals	Steroids	IL-Inhibitors	
Ebola virus disease	No study	Beneficial in a retrospective study; no effect on non-randomized clinical trials [37–39]	Remdesivir in clinical trial; some benefit [33]	No study	No study	mAb114, REGN-EB3 showed survival benefit in clinical trial [32]
Marburg virus disease	No study	No major clinical trial	None	No data	No data	Nothing approved, or on study
Crimean-Congo Hemorrhagic Fever	Observational studies and meta-analysis show benefit, especially when given early [81,82]	Compassionate use only, no RCTs	None	Potential benefit in severe cases [82]; no RCTs	No trials	Human mAbs under study; no clinical trials yet
Lassa fever	Observational studies show mortality benefit, especially when given early [65,66]	Small Phase-II study is conducted; results are pending [69]	None	No data	No data	Promising Arevirumab-3 cocktail, clinical data is needed [71]
Dengue fever	No study	No completed clinical trials; limited data	Celgosivir and balapiravir was tried in RCTs, no benefit [93]	Used in severe dengue; mixed evidence [97–99]	Not routinely used	No approved mAbs
Yellow Fever	No clear benefit	Not studied	None	No data	No data	Demonstrated good tolerance in phase I a/b clinical trial [118]
Rift Valley Fever	No clinical study	Not studied	None	No data	No data	No approved mAbs
Hantavirus	Effective in HFRS, especially when given early [132], not in HPS [134,135]	No data	None	Used in severe HPS; no controlled trials	No trials	No approved mAbs
Other Arenaviruses	No clinical study	No clinical study	None	No data	No data	No clinical mAbs

Abbreviations: RCT, Randomized Controlled Trial; mAb, monoclonal antibodies; HFRS, Hantavirus Fever Renal Syndrome; HPS, Hantavirus Pulmonary Syndrome.

6. Conclusion

In summary, oral antiviral drugs remain the cornerstone of treatment for most VHFs, alongside supportive care. Repurposing known antiviral drugs, especially during the early stages of infection, could lead to better outcomes. In the short term, it is critical that endemic countries improve their data collection on patient outcomes and treatment modalities. This will enable the development of robust observational studies, which can provide valuable insights. Monoclonal and neutralizing antibodies, along with other emerging therapeutic agents, continue to show strong potential across multiple VHFs; advancing these options will require the successful completion of ongoing phase trials, transparent reporting of results, and broader inclusion of endemic regions to ensure globally applicable and equitable treatment strategies.

7. Expert opinion

The advances in therapeutic strategies for VHFs discussed in this review have the potential to make significant real-world impacts, especially in improving disease management and patient survival. The availability of antiviral agents like ribavirin for CCHF and Lassa fever, alongside monoclonal antibodies for Ebola and Marburg virus, presents a new avenue for treatment that could reshape clinical practice. Early intervention could reduce mortality and improve prognosis, particularly when implemented together with improved diagnostic tools. However, the challenge lies in translating these therapeutic breakthroughs into real-world settings, where resource limitations, economic barriers, and logistical challenges remain significant obstacles to widespread adoption.

One of the major impediments in implementing these treatments lies in the practical limitations of healthcare infrastructure in endemic regions. Ribavirin, while effective, requires early administration and is not universally applicable to all VHFs, limiting its use. The monoclonal antibodies, which show great promise, are expensive and require specialized facilities for storage and administration, making them difficult to deploy in low-resource environments. These problems must be addressed to ensure that the advances in VHF therapies benefit populations in areas where these diseases are most prevalent. Overcoming these challenges will require international collaboration, economic investment, and innovative solutions for affordable production and distribution.

Improvement in diagnostic tools is another critical area for enhancing treatment outcomes. Early detection of VHFs is essential for initiating timely therapeutic interventions, but many of these diseases present with similar symptoms, making diagnosis challenging. Current diagnostic methods, such as PCR and serological assays, are often not feasible in resource-limited settings due to their cost and the need for specialized equipment and laboratory personnel. Developing rapid, cost-effective point-of-care diagnostic tools will be critical to improving early diagnosis, enabling healthcare providers to initiate appropriate treatment before the disease progresses to its later, more lethal stages.

Despite these challenges, the potential for further research in the field remains vast. Ongoing studies into novel antiviral agents, monoclonal antibodies, and immune-modulating therapies offer promising avenues for improving VHF treatment. The development of drugs targeting viral replication in early stages, along with therapies that modulate the host immune response in later

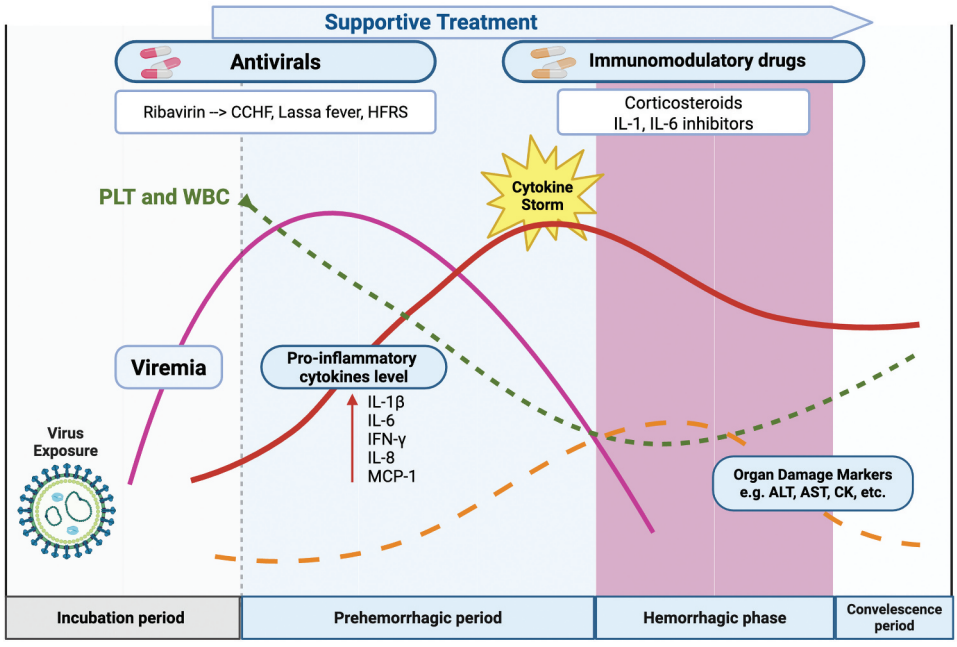


Figure 1. Conceptual overview of disease course and therapeutic windows in viral hemorrhagic fevers (VHFs). The schematic depicts the typical progression of VHFs from viral exposure through recovery, illustrating viremia, host immune activation, and the corresponding therapeutic phases. Antivirals are most effective during peak viremia in the pre-hemorrhagic phase, while immunomodulatory agents may help control hyperinflammation during the hemorrhagic phase. Rising cytokines and declining platelet/WBC counts mark the cytokine-storm stage, followed by elevated organ-damage markers and gradual recovery. The figure emphasizes the conceptual alignment between viral and immune dynamics and their respective therapeutic opportunities.

Abbreviations: CCHF, Crimean Congo Hemorrhagic Fever; HFRS, Hantavirus Fever Renal Syndrome; PLT, Platelets; WBC, White Blood Cell; ALT, Alanine aminotransferase; AST, aspartate aminotransferase. Created in BioRender. Güllü, D. (2025) https://BioRender.com/5tg2pet.

stages, could significantly improve patient outcomes. However, the path forward is not easy. Research in endemic regions faces limitations in funding and infrastructure, and the ethical concerns surrounding RCTs in high-mortality diseases complicate the design of studies. The lack of RCT-based evidence means that specific treatment guidelines for many VHFs remain absent. This highlights the importance of observational studies, which are increasingly relied upon to inform treatment approaches in the absence of RCTs.

In the upcoming years, the field of VHF research could evolve significantly, with substantial improvements in both prevention and treatment. As research advances, the availability of new, more targeted antiviral therapies could reduce the need for broad-spectrum treatments like ribavirin. Vaccine development, particularly for diseases like Marburg and Rift Valley fever, could help reduce the incidence of these diseases, leading to fewer outbreaks and a lower burden on healthcare systems. Additionally, the integration of real-time data collection and mobile health technology could further enhance disease surveillance and management, especially in remote areas, making it easier to track the spread of diseases and administer appropriate treatments. In the short term, repurposed antivirals may provide a practical and accessible option for inclusion in treatment guidelines, serving as an interim solution while novel therapeutics continue through development and evaluation.

In near future, we could see a significant shift in the standard approach to managing VHFs, with improved diagnostics, more effective and accessible treatments, and a broader use of vaccines. This evolution will require continued research, investment, and global collaboration to ensure that advancements in VHF treatment and prevention reach the populations most in need. While there are still many challenges to overcome, the future of VHF research is promising, and the next decade could bring transformative changes in the way these diseases are managed worldwide. The ongoing reliance on observational data, along with the potential integration of repurposed antivirals, will likely make treatment more adaptable and available in resource-limited settings, helping to shape more effective management strategies for VHFs.

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a foundational reference for optimizing supportive care in severe VHFs.

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